Detecting and Mitigating Algorithmic Bias in Credit Models: A Tutorial Using Public Financial Data

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***Abstract*—This tutorial presents a practical walk-through for detecting and mitigating algorithmic bias in credit scoring models using public financial datasets. We demonstrate the application of fairness metrics and mitigation techniques via open-source toolkits such as AIF360 and Fairlearn, with a focus on responsible and equitable AI in financial decision-making.**

***Index Terms*—Algorithmic Fairness, Credit Scoring, Bias Mitigation, Financial Data, Data-Centric AI, AIF360, Fairlearn**

I. Introduction

The adoption of automated decision-making systems in financial services has expanded rapidly, particularly in credit scoring and loan approval workflows. Machine learning models are increasingly used to assess creditworthiness based on historical applicant data. While these systems offer efficiency and scalability, they also inherit risks from the data on which they are trained. If sensitive or socially correlated features such as gender, age, or nationality are present—either directly or through proxies—the model may produce outcomes that disproportionately disadvantage certain groups.

This concern has brought algorithmic fairness to the forefront of AI governance and risk management. In credit decision contexts, unfair treatment can lead not only to reputational and ethical issues, but also to regulatory non-compliance. Traditional approaches to fairness in machine learning often focus on removing protected attributes from the dataset (fairness through unawareness), but this is insufficient when bias persists via correlated features. To address this, the field has developed a range of fairness-aware methods for auditing and mitigating bias throughout the model lifecycle. However, many practical implementations—especially in financial applications—lack accessible, reproducible workflows that illustrate these methods using real-world data.

II. Identified Gaps

This work aims to fill this gap by presenting a hands-on tutorial for auditing and mitigating algorithmic bias in a credit scoring scenario. Using the South German Credit dataset, we walk through the full modeling pipeline, from data preparation and sensitive attribute analysis to fairness evaluation and mitigation. Our goal is to provide a reproducible baseline for researchers and practitioners seeking to understand how fairness tools such as AIF360 can be applied in practice to financial decision models.

III. Theoretical Foundations

A. Algorithmic Bias: Definition and Origins

Algorithmic bias refers to systemic tendencies in AI models that result in unfair or unequal treatment of individuals or groups [1], [2]. These biases may arise unintentionally and often reflect historical inequalities embedded in training data or system design choices. In the context of credit scoring, for example, a model trained on past lending decisions may disproportionately reject applicants from minority backgrounds, perpetuating past discrimination.

Common sources of algorithmic bias include:

- **Biased training data:** Historical datasets often encode human prejudices or institutional inequalities [1].
- **Non-representative samples:** Poor sampling can underrepresent minority groups, skewing model behavior.
- **Optimization objectives:** Models that maximize overall accuracy may inadvertently harm minority subgroups [2].
- **Proxy variables:** Even when sensitive attributes like race or gender are removed, correlated features may leak this information [3].

B. Algorithmic Fairness: Group vs. Individual

Algorithmic fairness seeks to ensure that automated decisions treat people equitably. Two primary frameworks exist:

1) Group Fairness: Group fairness ensures that aggregated outcomes (e.g., approval rates, error rates) are similar across demographic groups [4], [2].

- **Demographic Parity:** Equal acceptance rates across groups.
- **Equal Opportunity:** Equal true positive rates among qualified individuals.
- **Equalized Odds:** Equal true and false positive rates across groups [4].

2) Individual Fairness: Introduced by Dwork et al. [5], individual fairness states that similar individuals should receive similar outcomes. This requires a well-defined similarity metric over input features. While conceptually appealing, operationalizing this metric can be challenging.

C. Fairness Metrics in Machine Learning

To measure and mitigate algorithmic bias, researchers have defined several mathematical metrics that capture different aspects of fairness. Below are the most widely used metrics in binary classification tasks such as credit approval.

1) Demographic Parity (Statistical Parity): A model satisfies **Demographic Parity** if the predicted positive rate is equal across all groups:

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b)$$

Where A is a sensitive attribute (e.g., gender or nationality). This metric ensures that members of all groups have an equal chance of receiving a favorable outcome.

Example: If 70% of male applicants are approved for credit, demographic parity would require that 70% of female applicants are approved too.

Limitation: This ignores whether individuals *deserve* the positive outcome (i.e., are qualified).

2) Equal Opportunity (EO): A model satisfies **Equal Opportunity** if the true positive rate (TPR) is equal across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = a) = P(\hat{Y} = 1 \mid Y = 1, A = b)$$

This ensures that among those who *actually qualify*, the probability of approval is the same regardless of group.

Example: If qualified female applicants are approved 90% of the time, then qualified male applicants should also be approved 90% of the time.

Use case: Suitable when we care most about fairness in access to opportunities for those who deserve them.

3) Equalized Odds (EOdds): **Equalized Odds** requires that both the true positive rate (TPR) and false positive rate (FPR) are equal across groups:

$$\begin{aligned} &P(\hat{Y} = y \mid Y = y, A = a) = \\ &P(\hat{Y} = y \mid Y = y, A = b), \quad \text{for } y \in \{0, 1\} \end{aligned}$$

Example: Both the likelihood of correctly approving a qualified person (TPR) and the likelihood of incorrectly approving an unqualified person (FPR) must be the same across groups.

Advantage: This is a stricter and more comprehensive fairness criterion.

Drawback: It may be hard to satisfy when base rates (e.g., true default rates) differ between groups.

4) Predictive Parity: **Predictive Parity** requires equal precision across groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = a) = P(Y = 1 \mid \hat{Y} = 1, A = b)$$

This ensures that among those predicted to succeed (e.g., repay credit), the success rate is equal for all groups.

5) Calibration: A classifier is **calibrated** if for individuals with the same predicted score, the probability of a positive outcome is the same across groups.

Example: If both men and women are assigned a credit score of 0.8, then both should have the same actual chance of repaying the loan.

Trade-offs Between Metrics

It is mathematically impossible to satisfy all fairness metrics at the same time unless the base rates (e.g., real-world approval or repayment rates) are equal across groups [4]. For example:

- A model can be well-calibrated and satisfy predictive parity, but violate demographic parity.
- Enforcing demographic parity may lower accuracy or increase false positives.
- Equal Opportunity focuses on fairness among the qualified, while Demographic Parity ignores ground truth.

Implication for Practice

Choosing the appropriate fairness metric depends on legal, ethical, and domain-specific considerations. In credit scoring, regulators may prioritize avoiding *disparate impact* (related to demographic parity), while practitioners may prefer equal opportunity to ensure fairness among qualified borrowers.

D. Sensitive Attributes and Proxies

Sensitive attributes include race, gender, age, and other legally protected or ethically sensitive categories [1]. Removing these attributes from the dataset does not eliminate bias, as proxy variables can leak correlated information [2].

Examples of proxies include:

- `ZIP code` as a proxy for race
- `Employment duration` as a proxy for age
- `Credit amount` as a proxy for gender roles

Identifying proxies involves statistical analysis, such as mutual information and correlation matrices.

E. Bias Mitigation Techniques

Bias mitigation techniques are categorized into three stages of the ML pipeline [3], [6]:

1) Pre-processing: These methods modify the dataset before training:

- **Reweighing:** Adjusts instance weights to equalize representation [7].
- **Massaging:** Alters labels to correct historical disparities.
- **Disparate Impact Remover:** Alters feature values to match distributions across groups [8].

2) In-processing: These techniques modify the training algorithm:

- **Fairness regularization:** Penalizes unfair predictions during optimization [9].
- **Fair constraints:** Adds fairness constraints (e.g., bounded disparity) [10].
- **Adversarial debiasing:** Trains a predictor and an adversary that tries to infer protected attributes.

3) Post-processing: Post-hoc methods adjust predictions without retraining:

- **Threshold adjustment:** Uses different decision thresholds per group.
- **Reject Option Classification:** Alters uncertain predictions to favor disadvantaged groups [7].
- **Output flipping:** Probabilistically changes outputs to equalize group outcomes [4].

F. Limitations of Fairness Through Unawareness

Fairness through unawareness assumes that not using sensitive attributes ensures fairness [11]. However, this is flawed because:

- Proxy features can still leak sensitive information.
- The model cannot monitor or correct disparities if it "ignores" group identity.
- Intentional corrective action (e.g., affirmative mitigation) requires knowledge of group membership.

Modern fairness frameworks encourage active detection and correction of bias rather than passive ignorance.

Affirmative Mitigation (Group-Aware Fairness): An important alternative to "fairness through unawareness" is **affirmative mitigation**, which explicitly incorporates sensitive group membership into the modeling pipeline to reduce historical or structural inequality [11], [4], [7].

Rather than ignoring sensitive attributes, this strategy uses them to:

- Monitor disparities during training and evaluation,
- Apply corrective mechanisms (e.g., reweighting, group-specific thresholds),
- Favor disadvantaged groups in borderline decisions.

This is especially relevant when the data reflects prior discrimination or when "neutral" decisions reproduce inequality [1]. Affirmative mitigation acknowledges that fairness may require different treatment to reach equitable outcomes — a principle rooted in *substantive equality*.

One widely used technique is **Reject Option Classification** [7], which favors the disadvantaged group only in prediction regions with low model confidence. This approach improves fairness without degrading overall accuracy and is supported in toolkits like AIF360.

Affirmative mitigation strategies are sometimes criticized for creating apparent disparities in treatment. However, they are often legally and ethically justified when aiming to correct unequal starting points [11].

IV. Materials and Methods

A. Environment Setup

To execute this tutorial we will use Python 3.12 [12].

1) Clone the repository from GitHub:

```
$ git clone https://github.com/
    JuliaWPereira/fairness
$ cd fairness
```

2) Create a virtual environment and install the requirements:

```
$ python -m venv .venv
# Mac / Linux
$ source .venv/bin/activate

# Windows CMD:
$ .venv\Scripts\activate.bat

# Windows PowerShell:
$.venv\Scripts\Activate.ps1
```

```
$ pip install "pip <24.1" # to work
    with fairness
$ pip install -r requirements.txt
```

We must use pip version lower than 24.1 since Fairness package is malformed at the moment of this tutorial creation (June of 2025).

We will use `pandas` package to deal with the dataset of South German credit Data; `fastparquet` will be used to keep type changes between notebooks, and since parquet is a better structure to datasets; `numpy` and `fairness` are used at the numerical exercises; `matplotlib` and `seaborn` are used for visualization purposes. `ydata-profiling` is used to profile the data and analyse correlations and distributions in a faster way; `ipykernel` is needed to run python notebooks.

B. Dataset

1) Origin and Description: We employ the South German Credit Data from the UCI Machine Learning Repository [13], which contains 1000 anonymized credit applications collected between 1973 and 1975. This dataset is a refined version of the classic Statlog German Credit Data [14], but fixing some inconsistencies and providing corrections about the background information [13].

2) Feature Types and Preprocessing: We start analysing the Dataset at the notebook `notebook/01_data_quality.ipynb`.

This dataset contains 14 categorical variables, 2 boolean variables, and 3 numerical variables. Since the columns are written in German, we first translate their names using the codetable provided in [13]. Several features are coded numerically, so we also use the codetable to translate the numbers to interpretable labels.

Please refer to Table I for personal and demographic information, to Table II for financial and employment status, to Table III for credit history and obligations, and to Table IV for loan application details.

3) Target Variable: The binary target variable, `credit_risk`, represents the bank's assessment of the applicant: `0 = bad credit` and `1 = good credit`. It is important to note that the dataset contains an oversampled number of bad credit cases—a common approach in domains with imbalanced outcomes. However, this means that the observed class distribution does not reflect the real-world prevalence of defaults, showed in Fig. 1.

4) Sensitive Attributes and Proxy Risk: We identify three features as sensitive for the purposes of fairness analysis:

- **age** – a continuous variable that may be subject to age-based discrimination;
- **foreign_worker** – a binary variable associated with nationality and potential racial bias;
- **personal_status_sex** – a categorical feature encoding both gender and marital status.

In addition to directly sensitive attributes, we must also consider *proxy attributes*: variables that are not explicitly

TABLE I
PERSONAL AND DEMOGRAPHIC INFORMATION

German Name	English Name	Values / Description
`alter`	age	Applicant's age in years
`famges`	personal_status_sex	1: male - divorced/separated; 2: female non-single or male single; 3: male - married/widowed; 4: female - single
`beruf`	job	1: unemployed/unskilled - non-resident; 2: unskilled - resident; 3: skilled employee/official; 4: manager/self-employed/highly qualified
`pers`	dependents	1: 3 or more; 2: 0 to 2
`telef`	telephone	1: no; 2: yes (under customer name)
`gastarb`	foreign_worker	1: yes; 2: no

TABLE II
FINANCIAL AND EMPLOYMENT STATUS

German Name	English Name	Values / Description
`laufkont`	checking_account_status	1: no checking account; 2: balance ¡ 0 DM; 3: 0–200 DM; 4: 200 DM or salary 1 year
`sparkont`	savings_account_balance	1: no/unknown; 2: ¡ 100 DM; 3: 100–499 DM; 4: 500–999 DM; 5: 1000 DM
`beszeit`	employment_duration	1: unemployed; 2: ¡ 1 yr; 3: 1–4 yrs; 4: 4–7 yrs; 5: 7 yrs
`verm`	property	1: no property; 2: car or other; 3: life insurance/savings; 4: real estate
`rate`	installment_rate	1: 35%; 2: 25–34%; 3: 20–24%; 4: ¡20%
`wohn`	housing	1: for free; 2: rent; 3: own
`wohnzeit`	residence_since	1: ¡1 yr; 2: 1–3 yrs; 3: 4–6 yrs; 4: 7 yrs

TABLE III
CREDIT HISTORY AND OBLIGATIONS

German Name	English Name	Values / Description
`moral`	credit_history	0: delayed repayment; 1: critical account; 2: no credits or all paid; 3: current credits paid; 4: all bank credits paid
`buerge`	guarantors	1: none; 2: co-applicant; 3: guarantor
`bishkred`	existing_credits_count	1: 1; 2: 2–3; 3: 4–5; 4: 6
`weitkred`	other_installment_plans	1: bank; 2: stores; 3: none

TABLE IV
LOAN APPLICATION DETAILS

German Name	English Name	Values / Description
`laufzeit`	duration_months	Duration of requested loan (in months)
`hoehe`	credit_amount	Requested loan amount (in Deutsche Marks)
`verw`	purpose	0: others; 1: car (new); 2: car (used); 3: furniture; 4: TV/radio; 5: appliances; 6: repairs; 7: education; 8: vacation; 9: retraining; 10: business
`kredit`	credit_risk	0: bad credit; 1: good credit

sensitive but may carry correlated information about protected characteristics [15]. For example, employment duration or housing status may reveal age-related patterns, while credit amount may indirectly relate to gender roles in financial responsibility. In Subsection **??**, we present an empirical analysis of such correlations to identify latent proxies.

V. TUTORIAL WORKFLOW

In this section, we outline the full pipeline used to audit and mitigate bias in the South German Credit dataset. The process is divided into seven distinct stages, each aimed at identifying, quantifying, and addressing potential algorithmic unfairness.

A. Pipeline Overview

The tutorial follows this sequence:

1) Data loading and initial preprocessing (already explained at the setup process);
2) Mapping sensitive and proxy attributes;
3) Exploratory Data Analysis (EDA): feature distribution, class balance, and sensitive attribute correlation;
4) Base model training: Logistic Regression and Decision Trees using Scikit-learn;
5) Fairness Evaluation: applying metrics such as Demographic Parity Difference and Equal Opportunity Difference using AIF360;
6) Bias Mitigation: using pre-processing (Reweighing)

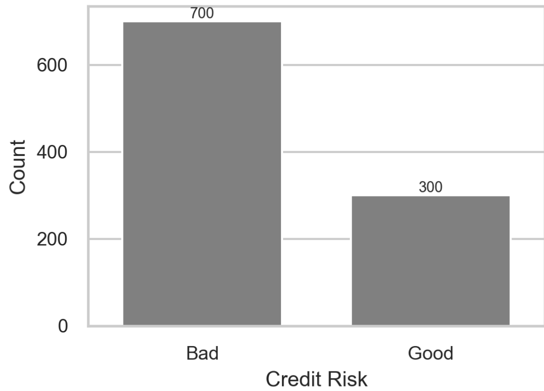

Fig. 1. Credit risk distribution (good vs bad).

and post-processing (Reject Option Classification) techniques;

7) Result Comparison: measuring accuracy and fairness before and after mitigation.

Each step is detailed in the following subsections.

B. Mapping Sensitive and Proxy Attributes

We define the following features as **sensitive attributes** for fairness evaluation:

- `age` – a continuous variable potentially subject to age-based discrimination;
- `foreign_worker` – a binary variable associated with nationality;
- `personal_status_sex` – a categorical feature combining gender and marital status.

To identify **proxy attributes**, we use two statistical approaches: *Pearson correlation*, to capture linear relationships, and *mutual information* (MI), to detect nonlinear dependencies. Since some features are categorical, we first encode them numerically.

```
from sklearn.preprocessing import LabelEncoder

df_encoded = df.copy()
for col in df_encoded.select_dtypes(include='
    object'):
     df_encoded[col] = LabelEncoder().
         fit_transform(df_encoded[col])
```

Listing 1. Encoding categorical features

We then assess which features leak information about `age` by computing correlations and mutual information with a binned version of age:

```
df_encoded['age_group'] = pd.qcut(df['age'], q
    =4, labels=False)

from sklearn.feature_selection import
    mutual_info_classif

X = df_encoded.drop(columns=['age', 'age_group
    '])
y = df_encoded['age_group']
mi = mutual_info_classif(X, y)
```

Listing 2. Analyzing proxy risk for age

At Table V we establish thresholds to understand which variables are really at risk to be proxy or just potential ones.

TABLE V
QUALITATIVE GUIDELINES FOR INTERPRETING MUTUAL INFORMATION (MI) SCORES IN PROXY DETECTION.

MI Score Range	Interpretation
0.00 – 0.02	Very weak / no dependency
0.02 – 0.07	Weak dependency (likely safe)
0.07 – 0.15	Moderate proxy potential
> 0.15	Strong proxy risk (flag feature)

Note: These thresholds are based on guidelines from feature selection heuristics [16], mutual information-based filtering for fairness [17], and practical usage in AIF360 [3].

The top correlated and informative features for `age` are shown in Figure 2. We observe that variables such as `employment_duration`, `residence_since`, and `credit_history` and `existing_credits_count`. This makes sense since older applicants tend to have longer employment history, longer time at a residence and more stable credit histories.

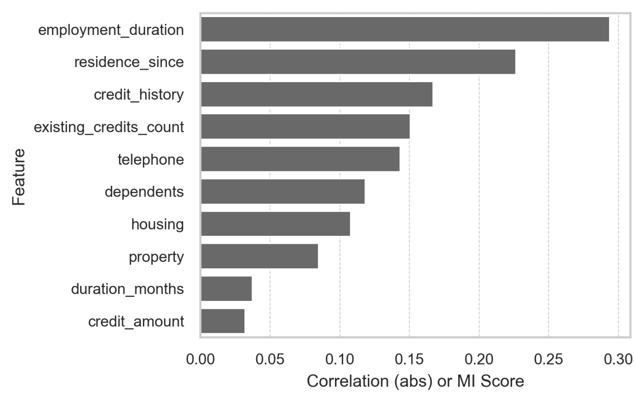

Fig. 2. Top correlated and informative features with `age`.

We repeat this process for `foreign_worker` and `personal_status_sex`, using the original categorical labels as targets.

For `foreign_worker`, there is no strong alert to a proxy, we have only two potential proxies with `duration_months` and `guarantors`, what makes sense with a more conservative approach in cases with foreigners which does not have a large credit history in the country to ask for guarantors.

In both cases, features such as `credit_amount`, `housing`, and `employment_duration` show high MI values, revealing indirect bias risks.

These results justify the need to monitor not only sensitive attributes but also proxy features that may introduce bias unintentionally. In our mitigation strategy, we either account

TABLE VI

FEATURES IDENTIFIED AS LIKELY OR POTENTIAL PROXIES FOR SENSITIVE ATTRIBUTES BASED ON MUTUAL INFORMATION. LIKELY PROXIES (MI > 0.15 FOR AT LEAST ONE ATTRIBUTE) ARE LISTED FIRST AND SHOWN IN **BOLD**.

Feature	Age	Foreign Worker	Personal Status / Sex	*Proxy Status*
employment_duration	**0.294**		0.063	*likely_proxy*
residence_since	**0.227**	0.045		*likely_proxy*
credit_history	**0.167**	0.039	0.102	*likely_proxy*
existing_credits_count	**0.151**		0.114	*likely_proxy*
dependents	0.119	0.077	**0.267**	*likely_proxy*
housing	0.108	0.033	**0.184**	*likely_proxy*
credit_amount	0.032		0.149	*potential_proxy*
guarantors		0.120		*potential_proxy*
duration_months	0.038	0.135	0.118	*potential_proxy*
telephone	0.144	0.075	0.087	*potential_proxy*

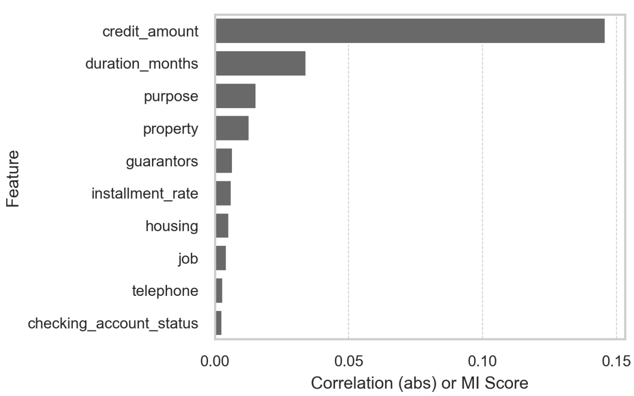

Fig. 3. Top features linked to `foreign_worker` status.

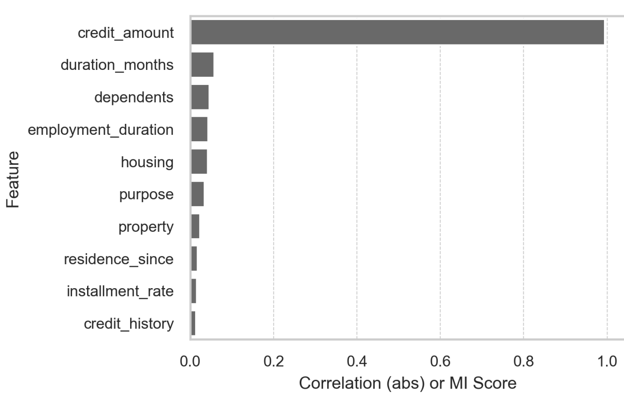

Fig. 4. Top features associated with `personal_status_sex`.

for or explicitly adjust these correlations to improve fairness outcomes.

1) Exploratory Data Analysis (EDA): We examined the distribution of each feature, checked for class balance, and visualized relationships between the target and sensitive attributes.

In Fig. 5 we can notice that the applicants with bad credit tend to be slightly younger on average in comparison with good credit risk ones. The bad curve also has a earlier around late 20s, while the good group is more spreaded. It suggests that age plays an implicit role in credit evaluation.

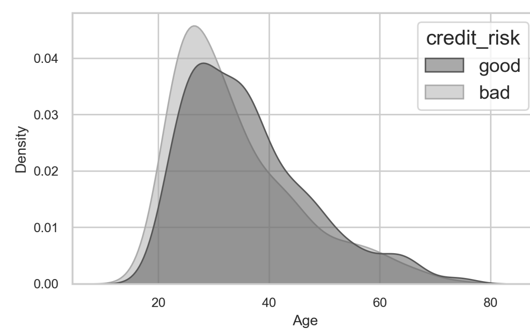

Fig. 5. Age vs. Credit Risk.

In Fig. 6 we can notice that the disparity between domestic applicants (`Foreign Worker == 'no'`) receiving **good** credit rating is high (667 good compared to 296 bad credit rating) while for foreigners it is considerably lower (33 good compared to 4 bad credit). Notice also that the foreigner group is under-represented (only 3.7% of the total data), limiting the model's ability to generalize fairly for this group, what amplifies the risk of discrimination.

For gender-status subgroups there are substantial differences. Male applicants, if married or widowed, receive the major number of approvals, followed by female applicants which are non-single or single male. In contrast, single female and divorced male have lower absolute counts and more balanced negative distribution. These differences suggest that marital status and gender may influence credit decisions. The underlying historical bias in these categories may result in unfair treatment, especially for under-represented groups such as single women or divorced men.

2) Baseline Model: Random Forest: Following the Statlog benchmark, we implemented a Random Forest classifier using Scikit-learn with 100 estimators and default hyperparameters. The dataset was split into training and test sets with a

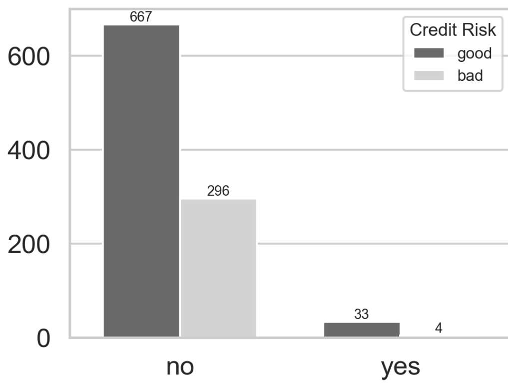

Fig. 6. Foreigner Worker vs. Credit Risk.

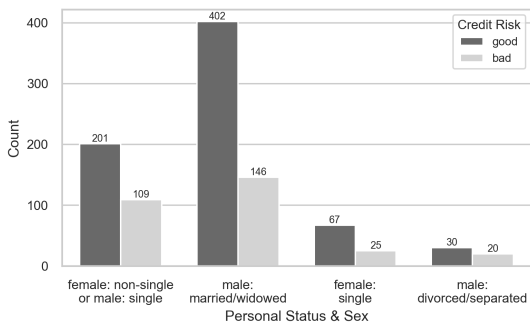

Fig. 7. Personal Status vs. Credit Risk.

70/30 ratio, and one-hot encoding was applied to categorical features. The sensitive attributes were excluded from training to simulate fairness through unawareness.

The model achieved an overall accuracy of 0.76. The classification report showed higher recall and precision for the positive class (good credit), but poor performance in identifying bad credit cases:

- **Accuracy:** 0.76
- **F1-score (class 0):** 0.48
- **F1-score (class 1):** 0.84

3) Fairness Evaluation: To evaluate fairness, we computed the *Demographic Parity Difference (DPD)* across three sensitive attributes: `foreign_worker`, `personal_status_sex`, and binned `age`. The results revealed the following disparities in selection rates between groups:

- **DPD (foreign_worker):** 0.2517
- **DPD (personal_status_sex):** 0.1512
- **DPD (age):** 0.0580

These disparities indicate that the model indirectly encodes bias through correlated features, even when sensitive variables are excluded. This motivates the need for mitigation techniques in the next stage.

4) Fairness Mitigation via Reweighing: To mitigate group-based disparities, we applied the Reweighing algorithm from the AIF360 toolkit [18]. This pre-processing method adjusts the instance weights to reduce bias in the training data, without altering the features or labels. We defined `foreign_worker` as the protected attribute, with "yes" as the privileged group.

We trained a Random Forest classifier on the reweighted dataset using the same architecture as before. The model achieved a higher overall accuracy of **0.839**, while significantly reducing the demographic disparity observed prior to mitigation.

- **Accuracy after reweighing:** 0.839
- **DPD after reweighing:** 0.0705

Compared to the baseline model (DPD = 0.2517), this result demonstrates that reweighing is effective in reducing bias while preserving or even improving predictive performance.

5) Result Comparison: Table VII summarizes the performance and fairness trade-offs between the baseline model and the mitigated model using reweighing. While the reweighted model improved overall accuracy, it also significantly reduced demographic disparities across all evaluated attributes.

TABLE VII
COMPARISON OF ACCURACY AND FAIRNESS BEFORE AND AFTER REWEIGHING

Model	**Accuracy**	**DPD (FW / PS / Age)**
Baseline RF	0.753	0.2517 / 0.1512 / 0.0580
Reweighed RF	0.839	0.0705 / 0.0814 / 0.0453

These results demonstrate that fairness-aware preprocessing can reduce bias across multiple dimensions while maintaining or even improving predictive performance.

VI. State of the Art

Recent advances in the literature on algorithmic fairness have refined and expanded the set of available strategies for bias mitigation in machine learning. Mitigation methods are typically classified into three categories: pre-processing, in-processing, and post-processing. Pre-processing methods focus on adjusting the dataset prior to training, aiming to neutralize unwanted correlations between sensitive attributes and the target variable. One prominent example is counterfactual data augmentation, where synthetic instances are generated with swapped sensitive attributes to reduce statistical dependence between protected variables and model outputs [19]. In-processing techniques incorporate fairness constraints directly into the learning process. A notable approach is representation neutralization, which modifies the internal representations learned by the model to obscure sensitive attributes, enhancing fairness without significant losses in predictive performance [20]. Post-processing methods operate after training, altering the final predictions to meet fairness criteria. These techniques are especially useful when access to training data is restricted or when model retraining is not feasible. Recent proposals include calibrated adjustment algorithms that re-score outputs to ensure fairness metrics such as demographic parity and equal opportunity are satisfied [21].

In the financial domain, particularly in credit scoring, the deployment of fairness-aware machine learning models is both a technical and regulatory challenge. Disparities in credit approval rates across demographic groups—such as age, gender, or nationality—may stem from historical bias encoded in training data. These disparities, when left unmitigated, not only compromise fairness but may also violate fair lending regulations [22]. Consequently, recent studies have evaluated fairness mitigation methods specifically in credit contexts, balancing predictive performance with ethical and legal considerations. For example, benchmark experiments with reweighing and reject option classification have demonstrated consistent reductions in demographic disparities without degrading model accuracy [23]. Furthermore, the literature has acknowledged the role of proxy variables, which are non-sensitive features that correlate with protected attributes and may inadvertently reintroduce bias even when direct sensitive attributes are excluded. As the use of AI models in financial decision-making expands, ensuring transparency and fairness through systematic auditing has become a central concern, especially for institutions operating under strict regulatory frameworks.

VII. Expected Contribution

This work provides a reproducible framework for fairness auditing in credit scoring models, using public data and open-source tools. It bridges fairness theory and applied machine learning by demonstrating concrete steps for bias identification and mitigation. Future work may explore alternative mitigation techniques or extend this approach to more complex financial contexts.

Acknowledgment

We acknowledge the authors and maintainers of AIF360 and Fairlearn for their tools and documentation.

Code Availability

The complete code and reproducible workflow used in this tutorial are available at:
https://github.com/JuliaWPereira/fairness.

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